FUNCTIONAL REQUIREMENTS SPECIFICATION

SaaS-Based Virtual School Device Management Platform

Market-ready functional scope divided into three independently sellable release phases

Document Type	Functional Requirements Specification (FRS)
Product Concept	Virtual school SaaS + MDM/kiosk platform for controlled learning devices
Primary Launch Platform	Android school-owned tablets using managed device enrollment and kiosk policies
Release Model	Three phases; each phase designed as a market-ready commercial product
Version	1.0
Date	May 17, 2026

Purpose: This document defines the functional product requirements for a market-ready virtual school device management platform. It is intended for prospective customers, investors, implementation partners, product teams, engineering teams, and solution architects.

Prepared for: prospective virtual school operators, edtech companies, device distributors, and enterprise implementation partners

1. Document Control

Item	Detail
Document title	Functional Requirements Specification - SaaS-Based Virtual School Device Management Platform
Version	1.0
Date	May 17, 2026
Intended audience	Prospective customers, product owners, engineering teams, implementation partners, sales teams, investors, and school leadership
Primary objective	Define the functional scope for a market-ready SaaS + MDM platform delivered in three independently sellable phases
Primary platform assumption	Phase 1 launches with Android school-owned managed devices; later phases add workflow depth and multi-OS enterprise support

2. Executive Summary

This document defines the functional requirements for a SaaS-based Virtual School Device Management Platform. The product enables schools to provide students with controlled learning devices that show only approved apps, websites, class resources, and learning modes. The product combines a cloud admin portal, Android MDM/kiosk enforcement, a student launcher, class scheduling, remote device support, reporting, and education-specific workflows.

The requirements are divided into three phases. Each phase is designed as a commercially useful product, not as an incomplete internal milestone. Phase 1 is a market-ready Android school-owned device control platform. Phase 2 adds education operations, teacher controls, parent engagement, exam workflows, BYOD study sessions, and integrations. Phase 3 expands into enterprise multi-OS management, partner/reseller operations, APIs, compliance workflows, and large-scale governance.

> **Market-ready principle:** Each phase must be sellable, supportable, deployable, documented, and able to deliver measurable value to a real school customer without depending on the next phase.

3. Product Vision and Scope

The product vision is to turn school-owned and approved student devices into controlled, distraction-free virtual classrooms. The system should help schools manage the entire device learning environment from a simple cloud dashboard rather than requiring complex enterprise IT expertise.

3.1 Product Objectives

- Allow schools to define exactly which apps, websites, and resources students may access during school activities.
- Provide scheduled learning modes such as Class Mode, Homework Mode, Break Mode, Exam Mode, Maintenance Mode, and Lost Device Mode.
- Enable remote enrollment, policy assignment, app deployment, device support, and compliance reporting.
- Give teachers practical live-class controls without giving them unsafe or excessive device privileges.
- Give parents appropriate visibility and support channels while preserving child privacy and school control.
- Scale from a single virtual school to multi-branch school networks, device distributors, and enterprise education groups.

3.2 In-Scope Product Components

Component	Description
Cloud SaaS admin platform	Tenant management, users, students, classes, policies, schedules, devices, reports, audit logs, billing usage, and support workflows.
Android management layer	Managed device enrollment, kiosk policy, app deployment, web restrictions, remote commands, compliance status, and device health.
Student launcher	A controlled student interface that shows only approved apps, resources, classes, messages, and support options.
Teacher console	Phase 2 feature for live class controls, session mode, pushed links, attendance, and class status.

Component	Description
Parent portal	Phase 2 feature for school-approved parent visibility, consent, lost-device reporting, and support.
Integrations	Phase 2 and 3 integration with LMS, SIS, Google Classroom, Microsoft education tools, assessment tools, webhooks, and partner APIs.
Enterprise/partner layer	Phase 3 features for multi-OS management, SSO, SCIM, reseller console, white-labeling, API platform, and compliance evidence.

3.3 Out-of-Scope or Restricted Capabilities

- The product should not be positioned as a covert surveillance tool.
- The product should not include hidden webcam monitoring, hidden microphone monitoring, keylogging, secret screenshots, or private message inspection.
- The product should not claim full device lockdown for student-owned BYOD devices unless the device is formally enrolled under a supported managed-device model with explicit consent and ownership controls.
- The product should not replace every existing enterprise MDM in Phase 1; it should launch as an education-specific managed Android device platform and expand carefully.
- The product should not treat unsupported platform features as guaranteed. Each OS and enrollment type must have a capability matrix.

4. Three-Phase Market-Ready Release Strategy

The release plan is organized around customer value. Each phase can be sold as a complete product for a defined customer segment, while also establishing a foundation for the next phase.

Phase	Market-ready product	Primary customer	Core value	Notable exclusions
Phase 1	Android Managed Learning Device Platform	Virtual schools issuing Android tablets to students	School-approved apps/websites only, scheduled learning modes, remote commands, inventory, and auditability	No full BYOD control, no multi-OS, no advanced LMS sync, no invasive monitoring
Phase 2	Learning Operations Platform	Virtual schools needing teacher, parent, exam, content, and integration workflows	Live class control, parent portal, exam mode, integrations, BYOD study sessions, category filtering, attendance and analytics	No government-scale multi-OS enterprise yet; BYOD remains session-limited
Phase 3	Enterprise Education Device & Learning Governance Platform	School chains, governments, device distributors, and enterprise education groups	Multi-OS abstraction, SSO/SCIM, partner console, APIs, compliance evidence, procurement lifecycle, enterprise analytics	Requires platform partnerships or API availability for some OS-specific enforcement

5. User Roles and Personas

Role	Primary responsibilities	Key system access
Platform Super Admin	Create tenants, manage platform settings, support enterprise accounts, monitor system health.	All tenants with strict platform audit controls.
School Owner/Admin	Own school configuration, users, branches, policies, reporting, and commercial settings.	Tenant-wide administration.
IT Admin	Enroll devices, create technical policies, manage app catalog, remote actions, device support.	Device, app, policy, command, and diagnostic modules.
Academic Admin	Manage classes, timetables, learning modes, exams, teacher assignments, academic reports.	Roster, schedule, exam, content, teacher workflow modules.
Teacher	Run live classes, push approved resources, monitor class readiness, mark attendance, create support requests.	Assigned classes only.
Student	Use controlled launcher, attend classes, access approved apps/resources, receive messages.	Student device app/launcher only.
Parent/Guardian	View approved status, consent, report lost device, manage school-approved after-school settings.	Linked children only.
Support Agent	Troubleshoot devices, handle tickets, communicate with admins/parents, record support notes.	Scoped support access.
Billing Admin	View plans, usage, invoices, renewals, and commercial limits.	Billing and usage views only.
Partner/Reseller Admin	Provision customer tenants, track implementations, support customers, view partner portfolio.	Phase 3 partner console.
Auditor/Privacy Officer	Review logs, evidence, consent, retention, incident records, and	Read-only governance and audit

Role	Primary responsibilities	Key system access
	access reviews.	access.

6. Functional Design Principles

Principle	Functional interpretation
Education-first	The product should organize device control around classes, grades, subjects, teachers, parents, and timetables rather than generic IT policies only.
Default safe access	For younger students, access should be default-deny: students see only approved apps/resources.
Policy transparency	Admins should always know why a device is in a particular mode and which rule caused it.
Child privacy	The system should control access without becoming a covert surveillance platform.
Teacher empowerment with guardrails	Teachers should be able to run class sessions but cannot bypass school-level safety restrictions.
Parent visibility, not override	Parents can help with support and after-school settings only within school-approved boundaries.
Platform-specific honesty	The product must display what is supported, partial, or unsupported by each OS and enrollment type.
Market-ready by phase	Every phase must include onboarding, support, auditability, documentation, and acceptance criteria.

7. Core Workflows

1. School onboarding: create tenant, configure school profile, school hours, roles, branches, privacy notices, and subscription limits.
2. Roster setup: create or import students, guardians, teachers, classes, grade levels, and teacher assignments.
3. Device enrollment: generate enrollment token/QR, enroll school-owned Android tablets, assign devices to students or classes, and verify compliance.
4. Policy creation: create baseline device policies, app catalogs, web allowlists/blocklists, launcher layout, and learning modes.
5. Schedule automation: configure timetables so devices automatically enter Class Mode, Homework Mode, Break Mode, or Maintenance Mode.
6. Student use: student opens the managed launcher and sees only approved apps, websites, resources, and messages for the active mode.
7. Remote support: IT admin reviews device status, triggers policy refresh, sends messages, locks lost devices, or opens support workflows.
8. Teacher live session: teacher starts class, pushes approved links/resources, and reviews session status (Phase 2).
9. Exam workflow: admin/teacher schedules exam, system performs readiness check, devices enter Exam Mode, and reports are generated (Phase 2).
10. Enterprise workflow: multi-OS policy simulation, SSO/SCIM, APIs, partner console, compliance evidence, and analytics (Phase 3).

8. Functional Data Model

The platform data model must support school operations, managed device policies, learning workflows, reporting, and auditability. The following are the minimum business objects required across the three phases.

Object	Purpose	Minimum attributes
Tenant	Represents a school, school group, or customer account.	Name, domain, timezone, country, plan, data region, status, branding, contacts.
Branch/Campus	Represents sub-organization under tenant.	Name, location, timezone override, contact, status.
User	Represents staff, teacher, parent, partner, or platform user.	Name, email/phone, role, status, tenant, branch, permissions, MFA state.
Student	Represents a learner.	Student ID, name, grade, class, branch, status, guardians, assigned devices.
Guardian	Represents parent/guardian linked to one or more students.	Name, contact, relationship, consent status, linked students.
Class/Section	Represents instructional group.	Class ID, grade, subject, teachers, students, timetable, policies.
Device	Represents a managed or tracked device.	Device ID, serial, OS, model, enrollment state, assigned student/class, policy, health.

Object	Purpose	Minimum attributes
Policy	Represents allowed apps, websites, restrictions, and launcher settings.	Name, version, status, assignments, rules, mode mapping, change history.
Learning Mode	Represents a contextual device state.	Mode name, start/end, allowed apps, websites, restrictions, priority, owner.
App	Represents approved app or web app.	Name, package/app ID, source, version, category, approval state, assignments.
Web Rule	Represents allow/block/category URL control.	Domain/URL/category, action, scope, source, expiry, priority.
Command	Represents remote device action.	Type, target, status, requester, timestamps, result, reason.
Event	Represents device, user, policy, or system event.	Type, actor, object, timestamp, metadata, severity.
Consent	Represents privacy/terms/parent consent acceptance.	Subject, actor, document version, date, method, status.
Ticket	Represents support request.	Requester, device/student, category, priority, status, notes, SLA.
Integration	Represents external system connection.	Provider, credentials reference, scopes, status, sync mappings, logs.

9. Policy and Mode Precedence

To avoid unpredictable device behavior, the platform must implement a deterministic policy resolution model. The active device state should always be explainable to an administrator.

Priority	Policy/mode type	Description
1	Lost Device Mode	Highest priority. Used for lost/stolen or emergency lockdown scenarios.
2	Exam Mode	Overrides normal timetable and teacher sessions during scheduled assessments.
3	Emergency/Safety Override	Temporary administrative override for urgent safety or access needs.
4	Teacher Live Session	Applies class-specific controls during a teacher-started session, within school guardrails.
5	Timetable Class Mode	Applies automatically based on class timetable.
6	Homework/After-school Mode	Applies outside school hours, including parent-configurable settings where allowed.
7	Device Baseline Policy	The default technical restrictions and app/web baseline for the managed device.
8	Default School Mode	Fallback state when no higher-priority mode applies.

Policy explanation requirement: Every device detail page must show the active mode, the source of the mode, the policy version, and the reason that policy won against other applicable rules.

10. Platform Capability Basis

The functional requirements rely on platform-supported device management capabilities. Android should be the first implementation target because dedicated device and kiosk-style capabilities are appropriate for school-owned learning tablets. Later phases can integrate with ChromeOS, Apple School Manager/MDM, and Microsoft Intune/Intune for Education where customer licensing and API access permit.

Platform	Functional basis	Phase
Android	Dedicated devices, kiosk or lock-task style restriction, app allowlisting, managed app deployment, kiosk customization, policy assignment.	Phase 1
ChromeOS	Managed guest sessions, kiosk/extension policies, Google Admin/ChromeOS device management concepts.	Phase 3
Apple/iPadOS	Apple School Manager, Automated Device Enrollment, supervised devices, app/content assignment, external MDM integration.	Phase 3
Windows	Microsoft Intune/Intune for Education, app/device management, compliance, remote actions, assessment-related workflows.	Phase 3
BYOD	Session-limited study mode or secure browser; no full personal-device takeover without proper managed-device enrollment and consent.	Phase 2

11. Phase 1 - Market-Ready Android Managed Learning Device Platform

Phase 1 delivers the first commercial product: a complete Android-first SaaS platform for school-owned devices. A school can onboard students, enroll devices, define approved apps/websites, run scheduled learning modes, manage devices remotely, and report on compliance. This is the minimum product that can be sold to virtual schools issuing Android tablets.

Area	Phase 1 market-ready expectation
Target customer	Virtual schools, online academies, coaching centers, and device-bundled education programs that issue Android tablets.
Commercial promise	Students can use only school-approved apps and websites during school-controlled modes.
Main buyer	School owner, IT/admin head, edtech operator, device distributor.
Deployment model	Cloud SaaS admin dashboard + Android managed device enrollment + student launcher.
Go-live requirement	A school can enroll devices, assign students/classes, apply policies, and begin controlled virtual learning without custom engineering.
Support readiness	Admin onboarding, device diagnostics, remote commands, audit logs, alerts, and basic help content.

11.1 Phase 1 Functional Requirements

Priority legend: M = mandatory for the phase to be market-ready; S = strongly recommended for the phase; C = optional/commercial enhancement.

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P1-FR-001	M	Tenant setup	The platform shall allow a platform super admin to create a new school tenant with school name, country, timezone, default language, logo, domain, plan, and primary contact.	A new tenant can be created from the platform console and is isolated from other tenants immediately after creation.	Super Admin
P1-FR-002	M	Tenant setup	The platform shall maintain strict tenant-level data isolation for users, devices, policies, events, billing records, and reports.	A user from one tenant cannot view, search, export, or manage another tenant's data unless explicitly assigned as a platform support user.	System
P1-FR-003	M	Tenant setup	The platform shall support school branches or campuses under one tenant.	Admins can create branches, assign users/devices/classes to a branch, and filter dashboards by branch.	School Admin
P1-FR-004	M	Tenant setup	The platform shall allow feature limits by subscription plan, including maximum devices, maximum students, available modules, and support tier.	Plan limits are visible to admins and enforced before exceeding configured limits.	Super Admin, Billing
P1-FR-005	M	Tenant setup	The platform shall provide a first-run setup checklist for new schools.	The checklist shows completion status for roster import, policy creation, app catalog setup, enrollment token generation, and first device enrollment.	School Admin
P1-FR-006	M	Tenant setup	The platform shall provide default policy templates for common virtual school scenarios.	Templates are available for Grade 1-3 lockdown, Grade 4-8 supervised learning, homework mode, exam starter mode, lost device mode, and maintenance mode.	School Admin, IT Admin
P1-FR-007	M	Tenant setup	The platform shall allow each tenant to configure school hours, weekend days, public holidays, and academic year dates.	Schedules and mode rules calculate based on the tenant timezone and calendar settings.	School Admin
P1-FR-008	M	Identity and RBAC	The platform shall support role-based access control for platform super admins, school admins, IT admins, academic admins, teachers, support agents, students, and parents.	Each role has a documented permission set and cannot access functions outside its assigned permissions.	System
P1-FR-009	M	Identity and RBAC	The platform shall allow admins to create, invite, suspend, and deactivate staff users.	An invited staff user receives an account activation link and can be deactivated without deleting historical actions.	School Admin
P1-FR-010	M	Identity and RBAC	The platform shall support secure login using email and password for admin and staff users.	Login includes password strength rules, lockout after repeated failed attempts, and password reset via verified email.	All Staff

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P1-FR-011	M	Identity and RBAC	The platform shall support multi-factor authentication for platform admins and school admins.	A tenant can require MFA for admin roles before allowing access to dashboard, policies, devices, and exports.	Super Admin, School Admin
P1-FR-012	M	Identity and RBAC	The platform shall allow custom permission assignment within predefined role boundaries.	A school admin can grant a teacher access to a class but cannot grant platform-level privileges.	School Admin
P1-FR-013	M	Identity and RBAC	The platform shall maintain session timeout and forced logout settings.	Admins can configure idle timeout; all sessions are invalidated after password reset or account suspension.	System
P1-FR-014	M	Identity and RBAC	The platform shall log all admin authentication events and privileged actions.	Audit logs show actor, timestamp, IP/device metadata, action, target object, and outcome.	System, Auditor
P1-FR-015	M	Roster	The platform shall allow admins to create and manage student profiles with student ID, name, grade, section, branch, status, and guardian links.	Student profiles can be created manually, edited, deactivated, and searched by name, ID, grade, and class.	School Admin
P1-FR-016	M	Roster	The platform shall support class, section, and grade management.	Admins can create classes, attach teachers, enroll students, and assign policies or schedules to those classes.	Academic Admin
P1-FR-017	M	Roster	The platform shall support bulk import of students, guardians, teachers, classes, and class membership using CSV files.	Imports validate required fields, show errors before commit, and provide downloadable error reports.	School Admin
P1-FR-018	M	Roster	The platform shall allow a student to be associated with one or more guardians.	Guardian records can be attached to multiple students where siblings exist.	School Admin
P1-FR-019	M	Roster	The platform shall support student statuses including active, pending, suspended, graduated, and withdrawn.	Inactive or withdrawn students cannot receive new device assignments unless reactivated.	School Admin
P1-FR-020	M	Roster	The platform shall allow teachers to view only the students and classes assigned to them.	Teachers cannot view unrelated classes unless granted access by an academic admin.	Teacher
P1-FR-021	S	Roster	The platform shall support academic-year rollover preparation.	Admins can duplicate or archive class structures and prepare next-year class assignments without deleting historical data.	Academic Admin
P1-FR-022	M	Device enrollment	The platform shall support Android managed device enrollment using enrollment tokens and QR-code based flows.	An IT admin can generate an enrollment token, enroll a supported Android device, and see it appear in inventory with status Online or Pending.	IT Admin
P1-FR-023	M	Device enrollment	The platform shall support bulk enrollment batches for school-owned devices.	An admin can create a batch, print or export enrollment QR codes, and track how many devices were enrolled under the batch.	IT Admin
P1-FR-024	M	Device enrollment	The platform shall require each managed device to be assigned to a tenant before it can receive policies.	Unassigned devices remain in a quarantine state and cannot access student learning modes until assigned.	System
P1-FR-025	M	Device inventory	The platform shall maintain device inventory records with serial number, model, manufacturer, Android version, device ID, enrollment date, assigned student/class, and policy version.	Device details are visible on a device profile page and exportable by authorized admins.	IT Admin
P1-FR-026	M	Device inventory	The platform shall show real-time or last-known device status including online, offline, non-compliant, locked, lost, retired, or enrollment failed.	Status changes update dashboards and device lists with timestamp of last check-in.	IT Admin
P1-FR-027	M	Device inventory	The platform shall support device grouping by branch, class, grade, device model, ownership type, and custom tags.	Policies and commands can be targeted using device groups and tags.	IT Admin
P1-FR-028	M	Device inventory	The platform shall allow reassignment of a device from one student to another with historical traceability.	The reassignment records old student, new student, actor, date, and reason.	IT Admin
P1-FR-029	M	Device inventory	The platform shall support device retirement and unenrollment workflows.	Retired devices stop receiving school policies and remain available in historical reports.	IT Admin
P1-FR-030	M	Device inventory	The platform shall detect duplicate serial numbers or duplicate enrollment attempts.	Admins are warned before duplicate device records are created.	System
P1-FR-031	M	Device inventory	The platform shall show device health indicators such as battery level, storage availability, OS version, last sync, app install status, and policy compliance.	A device detail page includes all health indicators with last refresh timestamp.	IT Admin
P1-FR-032	M	Android MDM	The platform shall allow admins to create baseline Android policies for school-owned managed devices.	A baseline policy can be created, named, versioned, and assigned to a device group.	IT Admin

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P1-FR-033	M	Android MDM	The platform shall support dedicated device kiosk mode for Android devices where platform capabilities allow it.	A device can boot into the selected kiosk launcher or kiosk application and prevent students from leaving the approved learning environment.	IT Admin, Student
P1-FR-034	M	Android MDM	The platform shall support multi-app kiosk behavior by presenting only approved apps inside the student launcher.	A student sees only the apps and web shortcuts allowed by the active policy and mode.	Student
P1-FR-035	M	Android MDM	The platform shall allow configuration of system navigation, status bar, notifications, and settings access according to platform-supported kiosk customization.	Admins can define whether system navigation, device settings, and notifications are available or hidden in each mode.	IT Admin
P1-FR-036	M	Android MDM	The platform shall prevent installation of unapproved applications where platform controls allow it.	Student cannot install apps from unapproved stores or sources on a fully managed school device.	Student
P1-FR-037	M	Android MDM	The platform shall allow operating system update windows and blackout periods to be configured for managed devices.	Admins can prevent disruptive updates during class/exam windows and allow them during approved maintenance windows.	IT Admin
P1-FR-038	M	Android MDM	The platform shall allow factory reset protection and reset restrictions where supported by Android version and device enrollment type.	Admins can enable reset restrictions; the UI clearly states platform and device limitations.	IT Admin
P1-FR-039	M	Android MDM	The platform shall maintain policy version history for every device policy.	Admins can see policy versions, change summary, actor, timestamp, and rollback to a previous version when allowed.	IT Admin
P1-FR-040	M	Android MDM	The platform shall support emergency unlock and support unlock workflows with reason capture.	A privileged admin can temporarily relax kiosk restrictions for support, and the action is logged with expiry time.	IT Admin, Support
P1-FR-041	M	Android MDM	The platform shall enforce a policy precedence model.	When multiple policies apply, the active policy is resolved using documented precedence: Lost Device > Exam > Teacher Override > Timetable Mode > Device Baseline > Default Mode.	System
P1-FR-042	M	App management	The platform shall provide an approved app catalog for each school tenant.	Admins can add app records with name, package ID, category, icon, description, approved grades, and installation rules.	IT Admin
P1-FR-043	M	App management	The platform shall allow admins to assign approved apps to policies, classes, grades, and device groups.	Assigned apps appear in the student launcher and unassigned apps do not appear.	IT Admin
P1-FR-044	M	App management	The platform shall support installation, update, and removal commands for approved Android applications where supported by the management API.	Admins can trigger or schedule app installation and view deployment status per device.	IT Admin
P1-FR-045	M	App management	The platform shall block or hide non-approved apps from the student learning environment.	Students cannot launch non-approved apps while school mode is active.	Student
P1-FR-046	M	App management	The platform shall display app install compliance status.	The dashboard shows installed, pending, failed, outdated, and blocked apps by device and class.	IT Admin
P1-FR-047	M	App management	The platform shall support web app shortcuts in the student launcher.	Admins can create a named web shortcut with URL, icon, description, and permitted grades/classes.	IT Admin
P1-FR-048	M	Web access	The platform shall support manual website allowlists and blocklists for managed browsing.	A website can be allowed or blocked at tenant, class, policy, or mode level.	IT Admin
P1-FR-049	M	Web access	The platform shall provide a secure learning browser or managed browser configuration for approved school websites.	Students can open approved websites and are blocked from non-approved websites according to policy.	Student
P1-FR-050	M	Web access	The platform shall allow admins to test a URL against a selected policy before publishing.	The tester shows Allowed, Blocked, or Unmatched with the rule that caused the decision.	IT Admin
P1-FR-051	M	Web access	The platform shall support a default-deny web model for younger grades.	When enabled, all websites are blocked unless explicitly allowed by the active policy or mode.	IT Admin
P1-FR-052	M	Web access	The platform shall record blocked web access events at an aggregated level without exposing unnecessary personal browsing details.	Reports show blocked event counts by policy, device, domain, and class, with privacy controls for detailed views.	IT Admin, Auditor
P1-FR-053	M	Learning modes	The platform shall support named learning modes including Class Mode, Homework Mode, Break Mode, Maintenance Mode, Lost Device Mode, and Default School Mode.	Each mode can define allowed apps, allowed websites, launcher layout, settings access, and notification rules.	School Admin, IT Admin

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P1-FR-054	M	Learning modes	The platform shall allow Class Mode to be scheduled by timetable.	Devices assigned to a class automatically enter Class Mode during scheduled class periods.	Academic Admin
P1-FR-055	M	Learning modes	The platform shall allow Homework Mode to run outside school hours with a different allowlist.	Homework Mode can allow homework tools while still blocking entertainment or unapproved apps.	School Admin
P1-FR-056	M	Learning modes	The platform shall allow Break Mode to relax selected restrictions during approved breaks.	Admins can configure which apps or websites are available during breaks and for how long.	School Admin
P1-FR-057	M	Learning modes	The platform shall allow Maintenance Mode for IT work.	Maintenance Mode can expose selected settings and support tools to authorized technicians only.	IT Admin
P1-FR-058	M	Scheduling	The platform shall support recurring timetables by class, grade, or student group.	Admins can define weekly schedules and assign modes to time blocks.	Academic Admin
P1-FR-059	M	Scheduling	The platform shall support schedule exceptions such as holidays, special events, exams, and emergency closures.	Exceptions override recurring schedules and are visible in a calendar view.	Academic Admin
P1-FR-060	M	Scheduling	The platform shall support manual mode override by authorized users.	An authorized admin can start, stop, or extend a mode and must enter a reason for audit logging.	School Admin, IT Admin
P1-FR-061	M	Remote actions	The platform shall allow authorized admins to remotely lock a managed device.	The device receives a lock command when online; if offline, the command is queued and executed on next check-in where supported.	IT Admin
P1-FR-062	M	Remote actions	The platform shall allow authorized admins to remotely unlock or release a device from lost/support state.	Unlock requires sufficient permission and creates an audit event.	IT Admin
P1-FR-063	M	Remote actions	The platform shall allow remote wipe or reset commands for lost or retired school-owned devices.	Wipe commands require confirmation, reason, and privileged permissions.	IT Admin
P1-FR-064	M	Remote actions	The platform shall allow admins to send a message to one device, a class, or a device group.	The message appears in the student launcher or as a managed notification depending on active policy.	School Admin, Teacher
P1-FR-065	M	Remote actions	The platform shall support request check-in or policy refresh commands.	Admins can force an online device to attempt policy sync and see updated result status.	IT Admin
P1-FR-066	M	Remote actions	The platform shall maintain a remote command queue and command history.	Every command shows submitted, queued, delivered, executed, failed, or expired status.	IT Admin
P1-FR-067	M	Support	The platform shall provide device diagnostics visible from the device detail page.	Diagnostics show latest policy, app deployment results, network status, battery, storage, and error messages.	Support Agent
P1-FR-068	S	Support	The platform shall provide a basic support case note feature attached to a device or student.	Support agents can add notes and change status without exposing unrelated student data.	Support Agent
P1-FR-069	M	Dashboards	The platform shall provide an admin dashboard summarizing devices, students, policies, app compliance, offline devices, and open alerts.	Dashboard numbers match detailed reports and can be filtered by branch, grade, class, and device group.	School Admin
P1-FR-070	M	Reports	The platform shall provide exportable device inventory reports.	Authorized admins can download CSV exports with device metadata and compliance status.	IT Admin
P1-FR-071	M	Reports	The platform shall provide policy deployment reports.	Reports show which devices received a policy, current policy version, deployment outcome, and error reason if failed.	IT Admin
P1-FR-072	M	Reports	The platform shall provide app deployment reports.	Reports show app installation success, pending status, failure, version mismatch, and affected devices.	IT Admin
P1-FR-073	M	Audit	The platform shall maintain immutable audit logs for critical actions.	Audit logs include actor, timestamp, action, object, old value/new value where applicable, IP/device metadata, and outcome.	Auditor
P1-FR-074	M	Notifications	The platform shall generate alerts for offline devices, failed policy application, failed app installation, lost device status, and abnormal repeated enrollment failures.	Alerts are visible in the dashboard and can be emailed to configured admin users.	IT Admin
P1-FR-075	M	Notifications	The platform shall allow tenant-level notification preferences.	Admins can configure alert recipients and alert thresholds for offline duration and failed deployments.	School Admin

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P1-FR-076	M	Privacy	The platform shall store privacy notices and acceptance records for school admins and guardians where parent access is used.	Consent records include version, acceptance date, subject, actor, and source channel.	School Admin
P1-FR-077	M	Privacy	The platform shall not provide covert webcam monitoring, covert microphone access, keylogging, or secret screenshots.	No Phase 1 user interface, API, or backend workflow enables these prohibited functions.	System
P1-FR-078	S	Privacy	The platform shall allow data retention settings for audit logs, device events, support notes, and inactive users.	Retention settings can be configured per tenant within platform limits and are documented in the tenant settings.	School Admin
P1-FR-079	M	Billing	The platform shall show tenant plan, usage, device count, student count, and upcoming renewal information.	Billing users can view usage and subscription details without access to student learning records.	Billing Admin
P1-FR-080	M	Onboarding	The platform shall provide in-product guidance for the first school setup and first device enrollment.	Admins can complete an end-to-end setup without vendor intervention using built-in steps, help links, and validation messages.	School Admin

12. Phase 2 - Market-Ready Learning Operations Platform

Phase 2 turns the device-control product into a complete virtual school operating layer. It adds teacher live-class controls, parent involvement, assessment mode, LMS/SIS integrations, category filtering, BYOD study sessions, content resources, attendance, analytics, helpdesk, localization, and school branding. Phase 2 can be sold as an upgrade to Phase 1 customers or as a higher-tier product for schools that need daily learning operations rather than device lockdown alone.

Area	Phase 2 market-ready expectation
Target customer	Virtual schools with active teachers, parents, live classes, assessments, LMS content, and support operations.
Commercial promise	The school can run controlled online classes and assessments while parents and teachers get practical workflows.
Main buyer	Virtual school operator, academic director, IT/admin head, parent-facing education provider.
Deployment model	Phase 1 Android core plus teacher console, parent portal, integrations, exam workflows, and BYOD study mode.
Go-live requirement	A school can run real class sessions, schedule exams, integrate class data, support parents, and measure engagement.
Support readiness	Ticketing, guided troubleshooting, integration health, parent support, and pilot feedback capture.

12.1 Phase 2 Functional Requirements

Priority legend: M = mandatory for the phase to be market-ready; S = strongly recommended for the phase; C = optional/commercial enhancement.

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P2-FR-001	M	Teacher console	The platform shall provide a teacher dashboard showing assigned classes, active sessions, today's timetable, student device status, and quick actions.	A teacher can open a class and see only assigned students and devices.	Teacher
P2-FR-002	M	Teacher console	The platform shall allow teachers to start a live class session that temporarily applies a class-specific mode.	When a session starts, assigned devices switch to the selected mode according to permissions and school policy.	Teacher
P2-FR-003	M	Teacher console	The platform shall allow teachers to push an approved app, website, LMS link, or lesson resource to students during a live session.	The pushed item appears on student devices and is limited to approved resources.	Teacher
P2-FR-004	M	Teacher console	The platform shall show student session presence, last check-in, mode status, and device readiness during class.	Teacher can identify students offline, not in class mode, or missing required apps.	Teacher
P2-FR-005	M	Teacher console	The platform shall provide an attention/focus screen function for managed devices.	A teacher can display an approved attention message or freeze the launcher for a short period according to school settings.	Teacher

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P2-FR-006	M	Teacher console	The platform shall allow temporary approval of a website during a class session.	Temporary approvals have expiry time, are limited to the class/session, and are logged.	Teacher
P2-FR-007	S	Teacher console	The platform shall allow teachers to save session templates for repeated classes.	Teachers can create reusable lesson modes within guardrails defined by admins.	Teacher
P2-FR-008	M	Teacher console	The platform shall keep a class session activity log.	Session logs include start/end, teacher actions, pushed resources, active mode, and major device status changes.	Teacher, Admin
P2-FR-009	M	Teacher console	The platform shall prevent teacher actions from overriding school-level restrictions unless explicitly allowed.	Teacher-level temporary actions cannot allow a blocked app/domain that school policy marks as never allowed.	System
P2-FR-010	M	Parent portal	The platform shall provide a guardian/parent portal for approved parent users.	Parents can log in and view only their linked children and school-approved information.	Parent
P2-FR-011	M	Parent portal	The platform shall show parents the student's school device status, school hours, upcoming mode schedule, and basic learning access status.	Parents can see whether the device is online/offline and what mode is expected without viewing sensitive classroom data.	Parent
P2-FR-012	M	Parent portal	The platform shall allow parents to report a lost or stolen school device.	A reported device creates an alert for school admins and can trigger Lost Device workflow if configured.	Parent
P2-FR-013	M	Parent portal	The platform shall allow school-approved after-school rules controlled by parents within school guardrails.	Parents may allow only apps/websites approved for parent control and cannot override school hours or exam modes.	Parent, School Admin
P2-FR-014	M	Parent portal	The platform shall support digital consent forms for guardians.	Parents can review and accept consent documents; acceptance is recorded with version and timestamp.	Parent
P2-FR-015	M	Parent portal	The platform shall provide parent notifications for device offline alerts, school messages, schedule changes, and support updates.	Parents can configure channel preferences where supported by tenant settings.	Parent
P2-FR-016	M	Parent portal	The platform shall allow parents to submit support requests related to device access.	Support requests are routed to school support or platform support based on tenant configuration.	Parent, Support
P2-FR-017	S	Parent portal	The platform shall provide multilingual parent-facing labels and messages.	Admins can configure default parent language per tenant or parent profile.	Parent, School Admin
P2-FR-018	M	Parent portal	The platform shall clearly separate parent visibility from surveillance.	Parent dashboards show approved summaries and do not expose private browsing history, messages, or camera/audio access.	System
P2-FR-019	M	Assessment	The platform shall provide an Exam Mode profile with strict app and website controls.	Exam Mode can limit devices to a test app or test domain and can hide all other launcher items.	Academic Admin
P2-FR-020	M	Assessment	The platform shall allow admins or teachers to schedule exams for a class, grade, or selected students.	Exam schedules override normal class/homework modes for the selected time window.	Academic Admin, Teacher
P2-FR-021	M	Assessment	The platform shall provide an exam readiness check before exam start.	The system identifies devices offline, missing test app, wrong policy version, low battery, or unsupported browser settings.	Teacher, IT Admin
P2-FR-022	M	Assessment	The platform shall support secure assessment launch links or exam apps.	Students enter the exam through an approved link or app; unapproved apps/websites are unavailable.	Student
P2-FR-023	M	Assessment	The platform shall restrict screenshots, copy/paste, notifications, and multitasking where supported by the platform and device type.	Restrictions are clearly marked as supported, partially supported, or not supported by each device type.	IT Admin
P2-FR-024	M	Assessment	The platform shall log exam integrity events without using invasive monitoring.	Logs may include mode exit attempts, policy violations, offline events, app launch failures, and support unlocks.	Teacher, Auditor
P2-FR-025	M	Assessment	The platform shall provide emergency exam unlock workflow.	Emergency unlock requires privileged permission, reason capture, and creates a high-severity audit event.	Academic Admin, IT Admin
P2-FR-026	M	Assessment	The platform shall release students from Exam Mode automatically at completion or at scheduled end time.	Release behavior follows exam configuration and does not leave devices stuck in exam mode.	System
P2-FR-027	M	Assessment	The platform shall provide exam session reports.	Reports include attendance, readiness, start/end time, integrity events, emergency unlocks, and device failures.	Academic Admin

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P2-FR-028	M	Integrations	The platform shall support Google Classroom integration for classes, rosters, teachers, coursework links, and materials where authorized.	An admin can connect Google Classroom, map courses to platform classes, and sync selected records.	School Admin
P2-FR-029	S	Integrations	The platform shall support Moodle integration for course links, user/course mapping, and LMS launch shortcuts.	Moodle courses can be linked to classes and displayed in the student launcher.	School Admin
P2-FR-030	S	Integrations	The platform shall support Canvas integration for course links and class mapping.	Canvas resources can be mapped to platform classes through configured API credentials.	School Admin
P2-FR-031	S	Integrations	The platform shall support Microsoft Teams or Microsoft 365 Education integration for class meetings and resources where authorized.	Teams meeting links and approved Microsoft learning apps can be pushed to class sessions.	School Admin
P2-FR-032	M	Integrations	The platform shall support Zoom and Google Meet meeting link management.	Teachers can push approved live-class links to students from the session console.	Teacher
P2-FR-033	M	Integrations	The platform shall support SIS/ERP imports through CSV and documented API endpoints.	Student, class, teacher, guardian, and enrollment data can be imported and reconciled.	School Admin
P2-FR-034	M	Integrations	The platform shall provide integration mapping tools.	Admins can map external course IDs, student IDs, teacher IDs, and class sections to platform records.	School Admin
P2-FR-035	M	Integrations	The platform shall show integration health, sync history, failures, and last successful sync.	Integration dashboard includes sync status, error reason, affected records, and retry options.	IT Admin
P2-FR-036	S	Integrations	The platform shall support outbound webhooks for key events.	Configured endpoints receive signed events for device offline, mode change, policy failure, exam started, support ticket created, and lost device reported.	IT Admin
P2-FR-037	M	BYOD study mode	The platform shall provide a BYOD Study Mode app or secure browser for student-owned devices.	Students can join a school-approved study session without the platform taking full control of their personal device.	Student
P2-FR-038	M	BYOD study mode	The platform shall require explicit student/guardian consent before using BYOD Study Mode.	Consent text explains what is controlled, what is collected, and what is not controlled on personal devices.	Student, Parent
P2-FR-039	M	BYOD study mode	The platform shall restrict BYOD control to the active study session only.	The platform cannot claim full device lockdown on BYOD devices and must clearly communicate limitations.	System
P2-FR-040	M	BYOD study mode	The platform shall provide minimum device/browser requirement checks for BYOD access.	Students receive a readiness result before class or exam access.	Student
P2-FR-041	M	BYOD study mode	The platform shall allow schools to decide whether BYOD participation is permitted per class, exam, and grade.	BYOD availability can be enabled or disabled by school policy.	School Admin
P2-FR-042	M	BYOD study mode	The platform shall record BYOD session attendance and participation status without collecting unnecessary personal-device data.	Reports show join time, session duration, approved resource access, and exit status.	Teacher, Auditor
P2-FR-043	M	Advanced web filtering	The platform shall add category-based filtering for common categories such as social media, games, adult content, proxies/VPNs, entertainment, file sharing, and education resources.	Admins can apply category rules at tenant, grade, class, and mode level.	IT Admin
P2-FR-044	M	Advanced web filtering	The platform shall support safe search and child-safe browsing settings where supported by the browser/search provider.	Admins can configure safe search enforcement and see coverage limitations.	IT Admin
P2-FR-045	M	Advanced web filtering	The platform shall support YouTube or video-resource allowlisting by channel, playlist, video, or approved educational domain where technically feasible.	Teachers can use approved educational video resources without opening unrestricted video access.	Teacher
P2-FR-046	M	Advanced web filtering	The platform shall allow global blocked domains and never-allowed rules.	Never-allowed rules cannot be bypassed by teacher temporary approval.	School Admin
P2-FR-047	M	Content library	The platform shall provide a school content library for approved apps, web links, PDF files, worksheets, LMS links, videos, and exam resources.	Resources can be categorized by grade, subject, class, and approval status.	Academic Admin
P2-FR-048	S	Content library	The platform shall support offline lesson packs for managed devices where supported.	Approved content can be cached on devices and shown in Offline Mode when connectivity is unavailable.	IT Admin, Student
P2-FR-049	M	Content library	The platform shall support content expiry and automatic removal.	Expired resources are hidden from student launcher and flagged in admin reports.	Academic Admin

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P2-FR-050	M	Attendance and analytics	The platform shall generate attendance based on device presence, class session join, LMS launch, or teacher confirmation depending on school configuration.	Attendance records can be reviewed, edited by authorized staff, and exported.	Teacher, Academic Admin
P2-FR-051	M	Attendance and analytics	The platform shall provide learning access analytics for admins and teachers.	Reports show class mode participation, app/resource launches, offline devices, and missed sessions.	Teacher, School Admin
P2-FR-052	M	Attendance and analytics	The platform shall provide device usage trend reports.	Admins can view usage by class, grade, device group, app category, and time window.	School Admin
P2-FR-053	S	Attendance and analytics	The platform shall provide low-engagement and device-risk alerts.	Rules can alert when a student repeatedly misses sessions, a device is frequently offline, or compliance repeatedly fails.	Teacher, School Admin
P2-FR-054	S	Attendance and analytics	The platform shall provide weekly summary reports for admins and optional parent summaries.	Reports can be generated and delivered based on tenant settings.	School Admin, Parent
P2-FR-055	M	Automation	The platform shall provide a rule engine for common device and learning operations.	Admins can create rules such as 'if device offline for X hours, notify IT and parent' or 'if policy fails twice, create support ticket'.	IT Admin
P2-FR-056	M	Support	The platform shall provide a helpdesk ticketing module.	Tickets can be created by admins, teachers, parents, or automated rules and linked to students/devices.	Support Agent
P2-FR-057	M	Support	The platform shall support remote assistance sessions where allowed and consented.	A support session requires authorization, displays notice, logs start/end, and does not silently monitor students.	Support Agent
P2-FR-058	S	Support	The platform shall provide device troubleshooting playbooks.	Common issues such as offline device, failed app install, blocked website, and stuck kiosk mode have guided resolution steps.	Support Agent
P2-FR-059	M	Branding	The platform shall support tenant branding for dashboard, launcher, parent portal, and email templates.	Schools can set logo, colors, support contact, and approved terminology.	School Admin
P2-FR-060	S	Branding	The platform shall support localization for core student, parent, and admin UI strings.	At least English plus one additional language can be configured for launch markets.	School Admin
P2-FR-061	M	Communications	The platform shall allow role-based announcements to students, parents, teachers, and admins.	Announcements can be targeted, scheduled, archived, and audited.	School Admin
P2-FR-062	S	Product feedback	The platform shall provide pilot feedback capture for teachers, parents, and admins.	Feedback can be collected by tenant, role, module, and severity.	Product Admin

13. Phase 3 - Market-Ready Enterprise Multi-OS and Ecosystem Platform

Phase 3 positions the product for enterprise-grade school groups, government education deployments, device distributors, and implementation partners. It adds multi-OS policy abstraction, ChromeOS/iPadOS/Windows integration paths, SSO, SCIM, partner/reseller console, public APIs, marketplace, compliance workflows, device procurement lifecycle, analytics, and operational governance.

Area	Phase 3 market-ready expectation
Target customer	School chains, government programs, regional education groups, enterprise virtual schools, device distributors, and implementation partners.
Commercial promise	One education-specific platform to govern controlled learning devices across OS families and organizational structures.
Main buyer	CIO/CTO, group academic director, procurement head, public-sector education team, reseller partner.
Deployment model	Enterprise SaaS with multi-OS integrations, identity integration, APIs, partner console, and governance center.
Go-live requirement	Enterprise tenants can connect identity systems, manage many schools, use partner workflows, and generate compliance evidence.
Support readiness	SLA dashboards, evidence packs, incident workflow, data residency options, partner implementation tracking, and enterprise analytics.

13.1 Phase 3 Functional Requirements

Priority legend: M = mandatory for the phase to be market-ready; S = strongly recommended for the phase; C = optional/commercial enhancement.

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P3-FR-001	M	Multi-OS policy	The platform shall provide a capability catalog that describes supported controls by OS, version, enrollment type, and device ownership model.	Admins can see whether a policy is supported on Android, ChromeOS, iPadOS, Windows, or BYOD before assignment.	IT Admin
P3-FR-002	M	Multi-OS policy	The platform shall provide cross-platform policy templates mapped to platform-specific implementation controls.	A school can define 'Grade 6 Class Mode' once and review generated platform-specific controls.	IT Admin
P3-FR-003	M	Multi-OS policy	The platform shall support platform-specific overrides within cross-platform policies.	An admin can set a different browser restriction for iPadOS than Android while keeping the same business policy.	IT Admin
P3-FR-004	M	Multi-OS policy	The platform shall provide policy simulation before rollout.	Admins can simulate how a policy will apply to selected devices and identify conflicts before publishing.	IT Admin
P3-FR-005	M	Multi-OS policy	The platform shall maintain multi-OS inventory in a unified device list.	Device list supports filtering by OS, ownership, enrollment type, management provider, compliance, and phase.	IT Admin
P3-FR-006	M	Multi-OS policy	The platform shall provide unified compliance scoring across OS families.	Compliance score considers platform-specific requirements and clearly explains missing or unsupported controls.	School Admin
P3-FR-007	M	Multi-OS policy	The platform shall support OS lifecycle and update governance.	Admins can identify unsupported OS versions and configure update recommendations or restrictions by platform.	IT Admin
P3-FR-008	M	ChromeOS	The platform shall integrate with Google Admin/ChromeOS management where authorized by the customer.	Admins can connect a Google Workspace domain and map ChromeOS organizational units to platform groups.	IT Admin
P3-FR-009	M	ChromeOS	The platform shall support ChromeOS kiosk app and managed session configuration mapping.	ChromeOS policies can be recommended or pushed through supported integration paths where available.	IT Admin
P3-FR-010	M	ChromeOS	The platform shall support managed guest session workflows for shared ChromeOS devices where licensed and configured by the school.	Shared devices can be linked to classroom or lab use cases without requiring a named student sign-in in every scenario.	IT Admin
P3-FR-011	M	ChromeOS	The platform shall support Chrome app and extension assignment reporting.	Admins can view required extensions/apps and deployment status for ChromeOS devices.	IT Admin
P3-FR-012	M	ChromeOS	The platform shall support ChromeOS web policy visibility.	URL allow/block settings and browser restrictions can be displayed in the platform policy view.	IT Admin
P3-FR-013	M	ChromeOS	The platform shall ingest ChromeOS inventory and status data where API access permits.	Device model, serial, OS version, last sync, assigned OU, and compliance status appear in unified inventory.	IT Admin
P3-FR-014	M	Apple/iPadOS	The platform shall support Apple School Manager integration planning and external MDM connection workflows.	Admins can configure Apple-related settings, tokens, and enrollment prerequisites with guided setup.	IT Admin
P3-FR-015	M	Apple/iPadOS	The platform shall support Automated Device Enrollment workflow visibility for school-owned iPad devices where connected to an MDM provider.	Admins can see supervised/enrolled status and device assignment state.	IT Admin
P3-FR-016	M	Apple/iPadOS	The platform shall support iPadOS supervised-device restriction mapping.	Policy view shows restrictions available for supervised iPad devices and flags unsupported controls.	IT Admin
P3-FR-017	M	Apple/iPadOS	The platform shall support app and book license assignment visibility where Apple School Manager and MDM integration allow.	Admins can see app availability, assignment, and installation status for iPad devices.	IT Admin
P3-FR-018	S	Apple/iPadOS	The platform shall support managed app configuration for supported education apps.	Admins can define app settings and assign them to iPad policies through supported MDM paths.	IT Admin
P3-FR-019	M	Apple/iPadOS	The platform shall track Apple push certificate and token expiry dates.	Admins receive alerts before certificates/tokens expire and risk interrupting device management.	IT Admin
P3-FR-020	M	Windows/Intune	The platform shall support Microsoft Intune or Intune for Education integration where authorized.	Admins can connect a Microsoft tenant and map groups/devices to platform classes or policies.	IT Admin

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P3-FR-021	M	Windows/Intune	The platform shall ingest Windows device inventory and compliance status where integration permits.	Windows devices appear in unified inventory with OS version, assigned user, compliance, and last sync.	IT Admin
P3-FR-022	M	Windows/Intune	The platform shall support Windows kiosk/assigned access policy mapping.	Admins can map learning modes to supported Windows kiosk or application restriction configurations.	IT Admin
P3-FR-023	S	Windows/Intune	The platform shall support assessment mode mapping for Windows where schools use Microsoft assessment tools such as Take a Test.	Exam policy view can include Windows-specific assessment configuration references and readiness checks.	Academic Admin
P3-FR-024	S	Windows/Intune	The platform shall support Autopilot or school PC deployment workflow visibility where available.	Admins can see deployment batches, assignment status, and failed provisioning indicators.	IT Admin
P3-FR-025	M	Windows/Intune	The platform shall show Windows application deployment status from the connected management provider.	Required apps show installed, pending, failed, and version status.	IT Admin
P3-FR-026	S	Windows/Intune	The platform shall ingest security and health reports where provider APIs permit.	Security, Defender, update, and configuration reports can be viewed or linked from the platform.	IT Admin
P3-FR-027	M	Enterprise identity	The platform shall support SAML 2.0 and OpenID Connect single sign-on for enterprise tenants.	Admins can configure SSO and require it for staff roles while preserving emergency break-glass accounts.	Enterprise Admin
P3-FR-028	M	Enterprise identity	The platform shall support SCIM-based user provisioning and deprovisioning.	Users and groups can be created, updated, suspended, and removed through identity-provider workflows.	Enterprise Admin
P3-FR-029	M	Enterprise admin	The platform shall support delegated administration across regions, branches, franchises, or school groups.	A regional admin can manage assigned schools but cannot access other regions.	Enterprise Admin
P3-FR-030	M	Enterprise admin	The platform shall support custom roles with permission templates.	Admins can create roles such as Device Operator, Exam Officer, Privacy Officer, and Implementation Partner.	Enterprise Admin
P3-FR-031	M	Enterprise admin	The platform shall support approval workflows for high-risk policy changes.	Policy changes such as broad unlock, web-block removal, or exam-mode change can require approval before publication.	Enterprise Admin, Auditor
P3-FR-032	M	Enterprise admin	The platform shall provide admin activity replay and change comparison.	Auditors can view what changed in a policy and who approved it.	Auditor
P3-FR-033	M	Enterprise admin	The platform shall support data residency configuration where offered by the platform.	Enterprise tenants can select supported hosting region during onboarding and see data-residency status.	Enterprise Admin
P3-FR-034	M	Enterprise admin	The platform shall support multi-region tenant structures for large school groups.	Reporting can roll up across regions while preserving local administration boundaries.	Enterprise Admin
P3-FR-035	M	API platform	The platform shall provide a documented public REST API for approved tenants and partners.	API documentation includes authentication, endpoints, rate limits, schemas, examples, and sandbox access.	Developer
P3-FR-036	M	API platform	The platform shall provide signed webhooks for critical events.	Webhook payloads include event ID, timestamp, tenant, object, type, signature, and retry behavior.	Developer
P3-FR-037	M	API platform	The platform shall support OAuth client credentials and scoped API keys.	API clients can be limited to specific permissions, tenants, and data scopes.	Developer, IT Admin
P3-FR-038	M	API platform	The platform shall expose API audit logs.	Admins can view API client, endpoint, action, timestamp, rate-limit status, and result.	Auditor
P3-FR-039	M	Marketplace	The platform shall provide an integration marketplace listing approved LMS, SIS, video, assessment, content, and support integrations.	Admins can enable, configure, test, and disable integrations from a marketplace page.	School Admin
P3-FR-040	S	Marketplace	The platform shall support partner-developed integration submissions subject to review.	Partner integrations can be reviewed, approved, rejected, and versioned by platform admins.	Partner, Super Admin
P3-FR-041	M	Partner/reseller	The platform shall provide a reseller console for implementation partners and device distributors.	Partners can view assigned customers, onboarding status, device counts, tickets, and renewal milestones.	Partner Admin
P3-FR-042	M	Partner/reseller	The platform shall allow partners to provision new customer tenants within approved commercial limits.	Partner-created tenants appear under partner portfolio and follow platform approval rules.	Partner Admin
P3-FR-043	M	Partner/reseller	The platform shall support white-label domains and branding packages for enterprise partners.	White-label configuration includes logo, colors, domain, support contacts, and email sender profiles.	Partner Admin

ID	Pri	Module	Functional requirement	Acceptance criteria	Primary roles
P3-FR-044	M	Partner/reseller	The platform shall support aggregated partner billing and usage reports.	Partners can see device/student counts, plan usage, renewals, and customer-level billing summaries.	Partner Billing
P3-FR-045	M	Partner/reseller	The platform shall support implementation project tracking.	Partner and customer teams can track setup milestones, risks, responsibilities, and go-live readiness.	Partner, Customer
P3-FR-046	M	Governance	The platform shall include a security and privacy center for each enterprise tenant.	Admins can view data settings, access controls, retention, audit exports, incident contacts, and policy documents.	Privacy Officer
P3-FR-047	M	Governance	The platform shall support downloadable compliance evidence packs.	Evidence packs include audit logs, access reviews, data-retention settings, security controls summary, and policy exports.	Auditor
P3-FR-048	M	Governance	The platform shall support data subject export and deletion workflows where legally and contractually applicable.	Authorized privacy users can request, approve, execute, and audit export/deletion workflows.	Privacy Officer
P3-FR-049	M	Governance	The platform shall provide incident management workflow for security or privacy incidents.	Incidents can be logged, classified, assigned, updated, closed, and exported for customers.	Security Admin
P3-FR-050	M	Governance	The platform shall provide periodic access review workflows.	Admins can review privileged users, confirm access, revoke access, and export evidence.	Enterprise Admin
P3-FR-051	M	Governance	The platform shall provide a privacy impact checklist for new tenant deployments.	Checklist covers data categories, purpose, consent, retention, support access, location use, and child-safety guardrails.	Privacy Officer
P3-FR-052	M	Operations	The platform shall expose service health and status dashboards to enterprise customers.	Customers can see incident status, maintenance windows, and affected modules.	Enterprise Admin
P3-FR-053	M	Operations	The platform shall support backup and restore operational controls for platform administrators.	Platform admins can verify backup status and perform controlled restore procedures according to runbooks.	Super Admin
P3-FR-054	M	Operations	The platform shall support disaster recovery readiness reporting.	Enterprise customers can receive RPO/RTO commitments and evidence according to contract tier.	Enterprise Admin
P3-FR-055	M	Device lifecycle	The platform shall support procurement batch tracking for bundled device + SaaS offerings.	Device batches can be tracked from purchase to enrollment, shipment, assignment, and replacement.	Operations
P3-FR-056	M	Device lifecycle	The platform shall support RMA and replacement workflows.	A damaged/lost device can be replaced while preserving student assignment history and support notes.	Operations, IT Admin
P3-FR-057	M	Analytics	The platform shall provide enterprise dashboards for regional performance, device compliance, attendance, engagement, and support trends.	Dashboards can roll up from school to branch, region, partner, and enterprise group.	Enterprise Admin
P3-FR-058	S	Analytics	The platform shall provide data warehouse export or scheduled report delivery for enterprise customers.	Reports can be exported to CSV/API storage destinations according to tenant configuration.	Enterprise Admin
P3-FR-059	S	Analytics	The platform shall provide policy recommendations based on compliance and usage patterns.	Recommendations are explainable suggestions and never change policies without approval.	School Admin
P3-FR-060	S	Analytics	The platform shall provide anomaly detection for unusual device, policy, or admin activity.	High-risk anomalies generate alerts and appear in the security center with supporting evidence.	Security Admin
P3-FR-061	S	Customer success	The platform shall generate quarterly business review packs for enterprise customers.	QBR packs include adoption, device health, policy success, support performance, learning mode usage, and recommendations.	Customer Success

14. Functional Acceptance Test Scenarios

The following acceptance scenarios should be included in product QA, pilot sign-off, and customer go-live validation. They are not exhaustive test cases, but they represent the highest-value market-readiness workflows.

Test ID	Workflow	Scenario	Expected result
P1-TC-001	Tenant onboarding	Create a tenant, configure school hours, create admin users, and complete setup checklist.	Tenant is active and ready for first device enrollment.
P1-TC-002	Roster import	Import 500 students, guardians, 30 teachers, and 40 classes from CSV with a deliberate error row.	Valid rows import, error row is reported, and no duplicate records are created.
P1-TC-003	Device enrollment	Enroll 25 Android tablets using QR codes and assign them to students.	All devices appear in inventory with accurate status, assigned student, and baseline policy.
P1-TC-004	Kiosk enforcement	Apply multi-app kiosk with LMS, video class app, notes app, and secure browser.	Student cannot open apps outside the allowlist or leave the managed launcher.
P1-TC-005	Class schedule	Create a timetable and trigger Class Mode automatically.	Devices enter and exit Class Mode at correct tenant-local times.
P1-TC-006	Web allowlist	Allow LMS and block non-approved entertainment domains.	Allowed websites open; blocked websites are denied and logged according to privacy settings.
P1-TC-007	Remote lock	Mark a device as lost and send remote lock/message command.	Device locks when online and displays the configured message.
P1-TC-008	Policy rollback	Publish a policy change, then roll back to previous version.	Devices receive the selected prior version and audit log records the rollback.
P2-TC-001	Teacher live session	Teacher starts a class session and pushes a lesson link.	Only assigned student devices receive the link and session activity is logged.
P2-TC-002	Parent lost device	Parent reports a device lost from portal.	School alert is created and lost-device workflow can be initiated by an authorized admin.
P2-TC-003	Exam mode	Schedule an exam and run readiness check.	Exam profile applies, unsupported devices are flagged, and post-exam report is generated.
P2-TC-004	Google Classroom sync	Connect Classroom and map courses/classes.	Classes, rosters, and coursework links sync according to authorization and mapping rules.
P2-TC-005	BYOD study session	Student joins a study session on personal device.	Access is session-limited, consent is recorded, and no full-device control is claimed.
P2-TC-006	Automation rule	Create rule: offline device > 24 hours triggers admin and parent alert.	Alerts fire when condition is met and stop after device check-in or closure.
P3-TC-001	Cross-platform policy	Create one Class Mode policy and map it to Android, ChromeOS, iPadOS, and Windows capabilities.	Policy simulation flags supported and unsupported controls per platform.
P3-TC-002	SSO/SCIM	Connect identity provider with SSO and user provisioning.	Users are provisioned, group membership maps to roles, and deprovisioned users lose access.
P3-TC-003	Partner tenant creation	Partner creates a new tenant under reseller console.	Tenant is created within partner permissions and appears in partner portfolio.
P3-TC-004	Compliance evidence	Privacy officer downloads evidence pack.	Pack includes relevant controls, logs, retention settings, and access review records.

15. Release Exit Criteria by Phase

Phase	Functional exit criteria	Operational exit criteria
Phase 1	Tenant, roster, Android enrollment, app/web policies, kiosk launcher, modes, scheduling, remote commands, reports, alerts, privacy guardrails, and audit logs are production ready.	Admin onboarding, help content, deployment checklist, support workflow, pilot runbook, and commercial usage limits are ready.
Phase 2	Teacher console, parent portal, exam mode, BYOD study mode, LMS/SIS integrations, category	Integration support process, parent-facing documentation, exam runbook, support SLAs, and

Phase	Functional exit criteria	Operational exit criteria
	filtering, content library, attendance, automation, helpdesk, and branding are production ready.	feedback workflows are ready.
Phase 3	Multi-OS capability matrix, enterprise identity, APIs, partner console, governance center, procurement lifecycle, analytics, and enterprise operations are production ready.	Security evidence pack, enterprise onboarding, partner training, status page, incident process, DR evidence, and account management workflows are ready.

16. Commercial Packaging Recommendation

The requirements support a tiered commercial model that aligns with the three phases. This allows the company to sell a focused product early while creating a clear upgrade path.

Package	Includes	Ideal customer
Core	Phase 1 Android managed devices, kiosk launcher, app/web allowlists, schedules, remote actions, inventory, reports, and audit logs.	Virtual school or academy issuing Android tablets.
Plus	Core + Phase 2 teacher console, parent portal, exam mode, integrations, BYOD study mode, category filtering, content, attendance, helpdesk, and branding.	Growing virtual school with live class operations and parent support needs.
Enterprise	Plus + Phase 3 multi-OS support, SSO/SCIM, APIs, partner console, governance, analytics, data residency, and enterprise operations.	Large school group, government program, or device distributor.

17. Key Assumptions and Dependencies

- Phase 1 enforcement assumes school-owned Android devices are enrolled under a supported managed-device model. Normal consumer apps cannot reliably control the whole device.
- Kiosk, app management, browser control, update control, reset control, and remote command behavior may vary by Android version, device manufacturer, enrollment type, and API support.
- BYOD study mode in Phase 2 is intentionally limited to session-based learning access and should not be sold as full personal-device lockdown.
- ChromeOS, iPadOS, and Windows requirements in Phase 3 depend on customer licensing, identity setup, API access, and external MDM/management-provider capabilities.
- Privacy, consent, data retention, and child-safety requirements must be localized for each target market before production deployment.

18. Privacy and Child-Safety Functional Guardrails

Guardrail	Requirement
No covert surveillance	The platform must not silently record webcam, microphone, keystrokes, personal messages, or private screenshots.
Purpose limitation	Data collection must be tied to device management, learning access, support, compliance, attendance, or safety workflows.
Role-limited access	Teachers, parents, support agents, and admins must see only the data necessary for their responsibilities.
Consent records	Guardian, school, staff, and BYOD consent must be versioned, auditable, and revocable where applicable.
Transparent support	Remote support and troubleshooting workflows should be visible, permissioned, and logged.
Retention controls	The system must allow retention configuration for logs, events, support notes, and inactive student records.
Export and deletion	Enterprise phase must support data export and deletion workflows where legally and contractually applicable.

19. Glossary

Term	Meaning
MDM	Mobile Device Management: technology used to configure, secure, and manage devices remotely.
Kiosk mode	A controlled device state where the user can access only one app or a limited set of approved apps.
Dedicated device	A fully managed device intended for a specific purpose, such as a school learning tablet.
Learning mode	A product-defined state such as Class Mode, Homework Mode, Exam Mode, Break Mode, or Lost Device Mode.
Allowlist	A list of apps, websites, or resources that are permitted.
Blocklist	A list of apps, websites, or resources that are denied.
BYOD	Bring Your Own Device. Student-owned or family-owned devices not issued by the school.
SIS	Student Information System used for official student, class, and enrollment records.
LMS	Learning Management System such as Moodle, Canvas, Google Classroom, or Microsoft Teams/Classroom-style tools.
SCIM	System for Cross-domain Identity Management used to provision and deprovision users from identity providers.

20. Reference Notes

The following public platform references informed the system capability assumptions in this requirements document. Final implementation must validate every capability against the target OS version, device model, enrollment type, customer license, and applicable API terms.

- **Android Management API - Dedicated devices and kiosk examples:** https://developers.google.com/android/management/policies/dedicated-devices
- **Android Enterprise - Dedicated devices overview:** https://developer.android.com/work/dpc/dedicated-devices
- **Android Enterprise - Lock task mode:** https://developer.android.com/work/dpc/dedicated-devices/lock-task-mode
- **Android Management API - Policies resource and kiosk customization:** https://developers.google.com/android/management/reference/rest/v1/enterprises.policies
- **Chrome Enterprise and Education - Managed guest session devices:** https://support.google.com/chrome/a/answer/3017014
- **Apple School Manager User Guide:** https://support.apple.com/guide/apple-school-manager/welcome/web
- **Apple School Manager - Automated Device Enrollment overview:** https://support.apple.com/guide/apple-school-manager/intro-to-apple-school-manager-axm7909096bf/web
- **Microsoft Intune for Education overview:** https://learn.microsoft.com/en-us/intune-education/what-is-intune-for-education
- **Microsoft Intune for Education device management overview:** https://learn.microsoft.com/en-us/intune/solutions/education/tutorial-school-deployment/manage-overview
- **Google Classroom API reference:** https://developers.google.com/workspace/classroom/reference/rest
- **Google Classroom API overview:** https://support.google.com/edu/classroom/answer/6253304
- **U.S. Department of Education - FERPA student privacy resources:** https://studentprivacy.ed.gov/ferpa

21. Requirement Count Summary

Phase	Mandatory	Strongly recommended	Optional/commercial enhancement	Total
Phase 1	77	3	0	80
Phase 2	50	12	0	62
Phase 3	52	9	0	61
Total	179	24	0	203

Document use: This FRS can be used as the basis for sales discussions, technical estimation, product roadmap planning, sprint epics, vendor evaluation, and customer pilot sign-off. It should be converted into detailed user stories and technical specifications before engineering execution.